

DATA CENTRE SYSTEMS ENGINEERING (M.TECH CSE)

Unit 2: Electrical Infrastructure Engineering

Easy-to-Understand Master Revision Deck for University Exams

Plain English Explanations • Simple Diagrams • High-Scoring Exam Tips • Solved Numericals

MIT-WPU × NIRVAA Solution Pvt. Ltd. | 100% Exam-Focused

The Journey of Electricity in a Data Centre (In Simple Words)

How power travels from government electric poles down to the CPU chip inside a server

The 6-Step Power Pipeline (Memorize This Flow!)

THE MASTER FLOW

- **1. High Voltage Grid (Substation):** Government power comes in at 33,000V or 11,000V through two different underground paths so one cable cut doesn't kill the building.
- **2. Step-Down Transformers:** Reduces the deadly 33,000V down to safe 415V/230V that equipment can use. Must be 'Dry-Type' (no oil fire risk).
- **3. Generator Backup (DG Sets):** If government power cuts, giant diesel engines automatically crank up and supply power within 10 seconds. They have enough diesel for 3 days!
- **4. The Battery Bridge (UPS):** While generators take 10 seconds to start, the UPS powers servers with ZERO milliseconds delay. The servers don't even flicker.
- **5. Floor Distribution (PDUs & Busway):** Overhead metal tracks on the ceiling drop power cables down to each server rack safely and cleanly.
- **6. Server Dual Power Supplies (Dual Cording):** Every server has two power cords (Plug A into Path A, Plug B into Path B). If one plug fails, the server never reboots!
- **Golden Exam Rule:** When asked for the Single Line Diagram (SLD), just draw these 6 blocks in order from top to bottom!

Utility Power: Why We Need Dual Feeds & RMU

Understanding how data centres connect to the power grid without ever losing power

Dual Utility Feeds (The 2-Road Rule)

GRID CONNECTION

- **The Problem:** If a data centre only connects to one electrical wire from the street, any road construction or vehicle crash cutting that wire shuts down Google/banking.
- **The Solution (Dual Feeds):** Bring in TWO separate power cables from TWO completely different power substations.
- **Physical Route Separation:** Cable A comes from North Street; Cable B comes from South Street (> 20 meters apart). If a backhoe digs up one road, the other line stays alive!
- **Ring Main Unit (RMU):** Think of RMU like a roundabout for electricity. If one road has an accident, the roundabout automatically routes power through the other road.
- **Gas-Insulated Switchgear (GIS):** Compact, sealed switch boxes filled with safe gas instead of open air. Takes up 70% less room in city buildings.

Transformers & K-Factor (Why Normal Transformers)

TRANSFORMERS

- **Cast Resin (Dry-Type) Transformers:** Built with solid epoxy resin instead of flammable oil. Safe to put directly inside the building. Cannot catch fire.
- **What are Harmonics?:** Normal appliances draw power smoothly like drinking water through a straw. Servers sip power in violent, jerky gulps (pulses). This is called 'Harmonics'.
- **The Overheating Danger:** These jerky pulses create high-frequency eddy currents that make normal transformers burn up and melt.
- **What is K-Factor?:** K-Factor is just a strength rating for transformers handling server electrical noise.
- **Standard Ratings to Remember:**
 - K-4: Basic office computers.
 - K-13: Standard server rooms.
 - K-20: Heavy cloud & AI data centres (handles 20x the eddy heat!).

ATS vs. STS: How Switches Transfer Power Without Crashing Servers

The differences in switching speed, technology, and placement in the data centre

ATS (Automatic Transfer Switch) - Mechanical

MECHANICAL SWITCH

- **How it works:** A physical motorized switch that flips between Source 1 and Source 2.
- **Transfer Speed:** Takes 50 to 100 milliseconds (0.05 to 0.1 seconds).
- **Can a server survive that?:** NO! Servers shut down if power is cut for more than 16 milliseconds (0.016s).
- **Where do we use it?:** Upstream! Between the Government Grid and the Diesel Generator. We don't care about the 100ms pause here because the UPS batteries downstream keep the servers running.
- **Wear & Tear:** Has mechanical moving parts that wear out over time.
- **Cost:** Affordable and very reliable for high-current switching.

STS (Static Transfer Switch) - Silicon Solid-State

ULTRA-FAST SILICON

- **How it works:** Uses computer chips called SCR thyristors. No moving parts at all.
- **Transfer Speed:** Blazing fast! Takes only 2 to 4 milliseconds (sub-cycle transfer).
- **Can a server survive that?:** YES! Servers have internal capacitor tanks that easily survive 16 to 20 ms. Servers don't even notice the switch!
- **Where do we use it?:** Downstream! Right near the server racks, switching between UPS Feed A and UPS Feed B.
- **Wear & Tear:** Zero moving parts; lasts forever with unlimited switches.
- **Exam Tip:** In exam tables, always state: ATS is slow (50-100ms) for Grid/DG; STS is ultra-fast (2-4ms) for Rack/IT!

Diesel Generators (DG Sets): The True Primary Power

Why data centres treat the government grid as a bonus and generators as the real backbone

Generator Ratings: What You Must Write in Exam

GENERATOR RATINGS

- **Standby Rating (ESP):** Only allowed to run for 200 hours per year at 70% load. Good for hospitals and apartments, but NOT allowed for Tier III/IV data centres!
- **DCC Rating (Data Centre Continuous):** The gold standard! Manufacturers certify these engines can run at 100% full power for 24/7/365 with ZERO hourly limits.
- **The 10-Second Black Start Rule:** When grid power cuts, generators must crank, fire, reach 50 Hz speed, synchronize, and accept 100% of the building load in less than 10 seconds (NFPA 110).
- **Jacket Water Heaters:** Electric heaters keep the generator engine block warm at 40°C all day so it starts instantly without choking on a cold winter night.
- **Dual Starter Batteries:** Two separate battery sets crank the engine. If Battery A is dead, Battery B starts it.

Fuel Tanks: How to Run for 3 Days Straight

72-HR FUEL STORAGE

- **Day Tank (Inside the Room):** Small fuel tank right next to the engine holding 1 to 2 hours of diesel for instant gravity feeding.
- **Bulk Storage Tank (Underground):** Huge tanks storing enough diesel fuel to run the data centre for 48 to 72 hours (3 full days) non-stop.
- **Fuel Polishing (Cleaning System):** Diesel sitting in tanks gets cloudy, grows bacteria, and collects water. Automated pumps filter and 'polish' the fuel 24/7 so it never clogs injectors.
- **Paralleling Switchgear:** When 4 big generators run together, this computer panel synchronizes their electrical waves like musicians playing in harmony so they don't blow up.
- **Exam Catchphrase:** 'Utility grid is an economic convenience; the on-site generator plant is the primary power source!'

Online Double Conversion UPS: The Zero-Downtime Heart

Why it's called 'Double Conversion' and how it achieves 0.00 milliseconds transfer time

ONLINE DOUBLE CONVERSION (VFI)

How It Works Step-by-Step (The 4 Building Blocks)

- **Block 1:** Rectifier (AC to DC) -> Takes dirty government wall power (Alternating Current) and turns it into clean Direct Current (DC). It also charges the battery bank.
- **Block 2:** Battery Bank (DC Storage) -> Sits permanently connected on the DC wire. Acts like a giant shock absorber and battery bank.
- **Block 3:** Inverter (DC to AC) -> Takes the clean DC power and converts it back into perfect, pure sine-wave 230V AC electricity for the server racks.
- **Why 'Double Conversion'?:** Because power is converted TWICE (AC -> DC -> AC). This completely cleans out all voltage spikes, surges, sags, and lightning noise!
- **Zero Milliseconds Transfer Time:** Because servers are ALWAYS drinking from the Inverter and the battery is ALREADY sitting on the DC line, when wall power fails, the transfer time is EXACTLY ZERO MILLISECONDS (0 ms). The servers don't even blink.
- **Static Bypass Switch:** If the UPS inverter has an internal electronic short-circuit, this ultra-fast chip switch instantly routes power around the UPS to wall power in 2ms so servers don't die.
- **Manual Maintenance Bypass:** A mechanical lever that lets an electrician turn off and safely replace the whole UPS box while servers stay powered.

Battery Showdown: Old Lead-Acid (VRLA) vs. Modern Lithium-Ion

Why modern cloud data centres are ditching heavy lead-acid batteries for smartphone-style lithium

VRLA (Old Lead-Acid Batteries)

TRADITIONAL (HEAVY)

- **What it is:** Heavy lead plates soaked in acid (just like a car battery, but sealed).
- **Short Lifespan:** Dies in only 3 to 5 years. Must be thrown out and replaced repeatedly.
- **Extremely Heavy:** Requires reinforced concrete floors (800-1000 kg/m²). Can crack regular office floors!
- **Takes Massive Space:** Requires huge dedicated rooms with special acid-proof ventilation.
- **Hates Heat:** Must be kept at exactly 20°C - 25°C. For every 10°C temperature rise, battery life is cut in HALF!
- **Low Initial Cost (Cheap):** Cheaper to buy on Day 1, but expensive over 10 years because you buy it 3 times.

Lithium-Ion (LFP Chemistry)

MODERN
(RECOMMENDED)

- **What it is:** Modern lithium chemistry (Lithium Iron Phosphate - LFP is fire-safe).
- **Long Lifespan:** Lasts 12 to 15 years (matches the whole lifespan of the data centre!).
- **60% Lighter & 70% Smaller:** Very light. Can be placed right inside the server room next to racks.
- **Handles Higher Heat:** Works happily up to 30°C. Saves thousands of dollars in air conditioning power!
- **Smart BMS (Brain):** Built-in Battery Management System monitors every cell's voltage, temperature, and health 24/7.
- **Cost:** Costs more on Day 1, but saves massive money over 10 years (Lowest Total Cost of Ownership).

Redundancy Topologies: N, N+1, and 2N Explained with Real Examples

How backup design determines whether a data centre is Tier I, II, III, or IV

N System (Zero Backup)

TIER I (BASIC)

- **Real-world Analogy:** A car with exactly 4 wheels and NO spare tire.
- **What it means:** If your data centre needs 1000 kW, you buy exactly one 1000 kW UPS.
- **The Risk:** If that one UPS breaks or needs an oil change, the whole data centre dies immediately.
- **Tier Rating:** Tier I (Basic capacity).
- **Availability:** 99.67% (allows 28.8 hours of downtime per year).

N+1 System (One Spare)

TIER II (SPARE COMPONENT)

- **Real-world Analogy:** A car with 4 wheels plus 1 spare tire in the trunk.
- **What it means:** You need 3 UPS units to run the servers? You buy 4 units (3 + 1 extra).
- **The Benefit:** If any 1 unit fails or needs maintenance, the other 3 carry the load.
- **The Flaw:** All 4 units connect to the SAME electrical pipe (single bus). If that bus faults, everything shuts down!
- **Tier Rating:** Tier II.

2N System (Two Completely Independent Paths)

TIER IV (FAULT TOLERANT)

- **Real-world Analogy:** An airplane with TWO completely independent engines and fuel lines.
- **What it means:** You build TWO separate power plants (Path A and Path B). Each can run 100% of the data centre alone!
- **Normal Operation:** Path A runs at 50%, Path B runs at 50%.
- **Zero Single Points of Failure:** If Path A blows up, catches fire, or gets struck by lightning, Path B runs everything smoothly.
- **Tier Rating:** Tier IV (The highest standard).

Power Distribution: Floor PDUs vs. Overhead Track Busways

Moving away from messy underfloor cables to clean, modular ceiling busbars

Old Way: Underfloor Cables (RPPs)

OLD METHOD

- **How it works:** Huge metal boxes called Remote Power Panels (RPP) stand on the floor. Electricians pull hundreds of thick cables under the raised floor tiles.
- **The Big Problem (Cable Spaghetti):** Hundreds of messy cables block cold air flowing under the floor. Servers overheat because cold air can't get through!
- **Zero Flexibility:** To add one new server rack, you must open floor tiles and pull a new heavy cable while other wires are live (danger of accidental unplugging).
- **Wastes Floor Space:** Bulky RPP panels take up room that could have held expensive server racks.

Modern Way: Overhead Track Busway

MODERN METHOD

- **How it works:** Metal electrical bars (tracks) are mounted on the ceiling directly above the rows of server racks.
- **Plug-In Tap-Off Boxes:** Little power boxes plug directly into the ceiling track anywhere you want, just like plugging into a wall socket!
- **Hot-Swappable:** An engineer can click in a new power box while the electricity is fully ON without cutting power to neighboring racks.
- **Free Cold Air:** Under the floor is 100% empty, allowing cold air to flow smoothly into servers (lowers electricity bills!).
- **Exam Tip:** Modern data centres use Overhead Busway because it saves floor space, improves cooling, and lets you add racks in minutes.

Inside the Server Rack: Intelligent ePDUs & Dual-Corded Servers

How redundant power reaches every single motherboard inside the rack

Rack PDUs (ePDUs) - From Dumb to Smart

RACK PDU TYPES

- **Basic PDU:** Just a dumb extension cord with multiple plugs. Distributes power, but has no screen and no network card.
- **Metered PDU:** Has a digital LED screen showing how many Amperes are being drawn. Prevents technicians from plugging in too many servers and tripping the breaker.
- **Monitored PDU:** Has an Ethernet port and IP address. Sends live voltage, power, and electricity usage alerts over the network to the control room.
- **Switched PDU (The Best!):** Lets engineers remotely turn individual outlets ON or OFF from home. If a server freezes at 2 AM, you can reboot it with one click on a web browser without driving to the data centre!

Dual-Corded Servers (Why Every Server Has 2 Plugs)

DUAL CORDING

- Every enterprise server (Dell, HP, Cisco) has TWO power supplies (PSU 1 and PSU 2) sticking out the back.
- **How they connect:**
 - Cord 1 plugs into Rack PDU A (powered by UPS Path A).
 - Cord 2 plugs into Rack PDU B (powered by UPS Path B).
- **Normal Operation:** Both power supplies share the work (50% each).
- **What happens during failure?:** If an electrician trips Breaker A or unplugs Cord 1 by mistake, PSU 2 instantly drinks 100% of the power from Cord B.
- **Result:** The website never goes down; users watching YouTube or banking never notice anything happened!

Earthing: Why the TN-S Scheme is Mandatory for Data Centres

Why keeping the Neutral and Earth wires separate saves human lives and stops data corruption

The 3 Reasons We Do Earthing

WHY GROUNDING MATTERS

- **1. Human Life Safety:** If a wire comes loose inside a metal server chassis, the current must rush harmlessly down to earth and trip the breaker in milliseconds so technicians don't get electrocuted.
- **2. Neutral Voltage Stabilization:** Ties the transformer center point to Earth so voltage doesn't float dangerously between 0V and 500V.
- **3. High-Frequency Noise Drainage:** Microchips run on tiny 1.2V signals. Static electricity and radio noise must be drained away so binary 1s don't flip to 0s!
- **Earth Resistance Rule:** Normal buildings allow 5 Ohms. Data centres demand less than 1.0 Ohm (target < 0.5 Ohm) using maintenance-free chemical copper earth pits.

Why TN-S is Mandatory (And TN-C is Forbidden)

TN-S IS MANDATORY

- **What is TN-S?:** 'Terre-Neutre Séparé'. It means the Neutral wire (N) and Earth wire (PE) are kept STRICTLY SEPARATE throughout the whole building.
- **The Only Bond:** Neutral and Earth are tied together at ONLY ONE SPOT (at the transformer). Never again downstream!
- **Why TN-C is Forbidden (Dangerous!):**
 - In TN-C, Neutral and Earth share one single wire (PEN wire).
 - Thousands of server power supplies dump electrical noise onto the neutral.
 - That noise flows through the metal cabinets and Ethernet shields!
 - Result: Electric shocks to workers and corrupted data packets.

Signal Reference Grid (SRG) & Lightning Protection

How a simple 2ft x 2ft copper mesh under the floor stops high-frequency computer noise

Signal Reference Grid (SRG) - In Plain Words

THE 2x2 FT MESH

- **The Problem with Normal Ground Wires:** At regular 50 Hz power, a long copper wire is a great ground. But at high frequencies (computer clocks in MHz), a single wire acts like an electrical wall (high inductive impedance) and radiates noise like a radio antenna!
- **The Solution (SRG Mesh):** A prefabricated 2 ft × 2 ft copper grid laid under the raised floor tiles.
- **The Parallel Wire Magic:** When you connect hundreds of wires in a grid, the electrical resistance/inductance divides down to near ZERO!
- **How to Connect:** Every server rack, cable tray, and floor pedestal is strapped to the nearest square on the copper grid with a short braided flat strap.
- **Result:** All servers share the exact same clean electrical floor. No ground loops; zero static ESD.

Lightning & Surge Protection (3-Stage Defense)

3-STAGE DEFENSE

- **External Lightning (Roof):** Metal rods on the roof and a wire mesh catch direct lightning strikes and safely guide thousands of Amperes down to deep earth pits.
- **Internal Lightning (The 3 SPDs - Surge Protectors):**
 - Type 1 SPD (Main Gate): Catches the massive direct lightning strike at the main building entrance switchboard.
 - Type 2 SPD (Floor Panels): Catches leftover surges at the floor distribution and UPS boards.
 - Type 3 SPD (In the Rack): Catches tiny residual spikes inside the rack PDU, clamping voltage down to safe levels (< 1000V).
- **Exam Rule:** SPDs must be cascaded in 3 stages: Heavy at the gate, Medium at the floor, Fine at the server!

How to Size a Data Centre: The 6-Step Calculation Pipeline

A simple step-by-step method to solve any university exam sizing question

Step-by-Step Sizing Logic (Follow This in Exams!)

THE SIZING RECIPE

- **Step 1:** Calculate Server Power (IT Load in kW) -> Multiply number of racks by kW per rack. Example: 100 racks $\times$ 5 kW = 500 kW.
- **Step 2:** Convert kW to kVA -> Real power (kW) divided by Power Factor (usually 0.95 or 0.98). Example: 500 kW / 0.95 = 526 kVA.
- **Step 3:** Size the UPS (kVA) -> Divide by UPS efficiency (e.g. 95%) and divide by maximum safe load factor (e.g. 75%). Example: 526 / (0.95 $\times$ 0.75) = 738 kVA -> Choose two 500 kVA units in N+1!
- **Step 4:** Size the Batteries (Ah) -> Calculate DC discharge current = DC Watts / DC Bus Voltage. Multiply current by backup hours (15 mins = 0.25 hrs). Multiply by 1.25 safety factor.
- **Step 5:** Calculate Total Building Power using PUE -> Total Power = IT kW $\times$ PUE. If PUE is 1.4: 500 kW $\times$ 1.4 = 700 kW (200 kW goes to AC cooling and fans).
- **Step 6:** Size the Substation Transformer & Generator (kVA) -> Total Building kVA + 20% buffer for future growth.
- **Exam Secret:** Examiners grade on whether you included power factor, UPS efficiency, and the 20-25% safety buffer!

PUE and DCiE: The Energy Efficiency Formulas Demystified

Understanding how much electricity does real work versus how much is wasted on cooling

PUE (Power Usage Effectiveness)

PUE FORMULA

- **The Formula:** $PUE = \text{Total Facility Power} / \text{IT Equipment Power}$.
- **The Analogy:** Imagine you buy 100 kg of food. If your family eats 100 kg and wastes 0 kg, your ratio is 1.0 (Perfect!). If you throw 50 kg in the trash, your ratio is 1.5.
- **What does PUE = 1.0 mean?:** Absolute perfection! 100% of the electricity goes into powering computer servers; zero electricity wasted on cooling or lights.
- **What does PUE = 2.0 mean?:** Terrible! For every 1 Watt used to power a server, you waste another 1 Watt running giant air conditioners.
- **Real-World Numbers:**
 - Old Data Centres: PUE = 1.8 to 2.0 (50% wasted!)
 - Modern Data Centres: PUE = 1.3 to 1.4
 - Google / Hyperscale Cloud: PUE = 1.10 to 1.15

DCiE (The Reverse Percentage Metric)

DCiE PERCENTAGE

- **The Formula:** $DCiE = (\text{IT Power} / \text{Total Power}) \times 100\% = (1 / PUE) \times 100\%$.
- **What it tells you:** It shows the efficiency percentage directly.
- **Quick Calculation Examples:**
 - If PUE is 2.0: $DCiE = 1 / 2.0 = 50\%$ (Only half your power does real work).
 - If PUE is 1.33: $DCiE = 1 / 1.33 = 75\%$.
 - If PUE is 1.10: $DCiE = 1 / 1.10 = 91\%$ (Hyperscale efficiency!).
- **How to Lower PUE in Exams (Mention These 3!):**
 1. Hot/Cold Aisle Containment (stops hot and cold air mixing).
 2. High-efficiency UPS units and transformers.
 3. Overhead busways (removes underfloor cable airflow blockage).

Uptime Institute Tier Standards: The 4 Tiers in Plain English

From a basic server room to a completely bulletproof Tier IV financial vault

Tier I: Basic

TIER I

- **Paths:** 1 Single Path.
- **Backup:** N (No spares).
- **Can you service it live?:** NO.
Turning off a switch turns off servers.
- **Downtime Allowed:** 28.8 hours / year.
- **Availability:** 99.671%.
- **Analogy:** A bicycle.

Tier II: Spare

TIER II

- **Paths:** 1 Single Path.
- **Backup:** N+1 spare components.
- **Can you service it live?:** Still NO, because the single power pipe cannot be turned off.
- **Downtime Allowed:** 22.0 hours / year.
- **Availability:** 99.741%.
- **Analogy:** Car with spare tire.

Tier III: Con

TIER III

- **Paths:** 2 Paths (1 Active, 1 Standby).
- **Backup:** N+1 modules.
- **Can you service it live?:** YES! You can unplug any pipe or switch for maintenance with ZERO IT downtime.
- **Downtime Allowed:** 1.6 hours / year.
- **Availability:** 99.982%.
- **Analogy:** Train with backup rail.

Tier IV: Fau

TIER IV

- **Paths:** 2 Simultaneously Active Paths.
- **Backup:** 2N or 2(N+1).
- **Can you service it live?:** YES!
- **Survives Accidents?:** YES! If Path A catches fire, Path B handles it automatically.
- **Downtime Allowed:** 26 minutes / year.
- **Availability:** 99.995%.
- **Analogy:** Dual-engine space jet.

Exam Problem 1: Sizing the UPS and Battery (Step-by-Step)

How to easily solve this guaranteed 10-mark university question without getting confused

Problem Statement & Clean Solution Walkthrough

EASY 10 MARKS

• **The Question:** A data centre has 500 kW IT load with 0.95 Power Factor. The UPS inverter is 94% efficient. Sizing rules state normal load must not exceed 75% of UPS rating. The battery DC bus is 480V, and it must provide 15 minutes of backup. Find: (a) IT kVA, (b) Minimum UPS kVA, (c) Battery Ah capacity.

• **Step 1:** Find IT Apparent Power (kVA)

Formula: $\text{kVA} = \text{kW} / \text{Power Factor}$

$\text{kVA} = 500 / 0.95 = 526.3 \text{ kVA}$. (Write: 'Total IT Apparent Power = 526.3 kVA').

• **Step 2:** Find the UPS Rating in kVA

Because the inverter is 94% efficient, it needs: $526.3 / 0.94 = 559.9 \text{ kVA}$.

Because of the 75% safe load rule, divide by 0.75: $559.9 / 0.75 = 746.5 \text{ kVA}$.

Conclusion: Specify two 500 kVA units in N+1 configuration (1000 kVA total).

• **Step 3:** Find the Battery Ah Capacity

DC Power needed = $500 \text{ kW} / 0.94 = 531.9 \text{ kW} = 531,900 \text{ Watts}$.

Discharge Current (Amps) = $\text{Watts} / \text{DC Volts} = 531,900 / 480\text{V} = 1,108 \text{ Amps}$.

For 15 minutes (0.25 hours): $1,108 \text{ Amps} \times 0.25 \text{ hours} = 277 \text{ Ah}$.

Add 25% safety buffer for battery aging (multiply by 1.25): $277 \times 1.25 = 346.25 \text{ Ah}$.

Final Answer: Choose a 480V DC battery bank rated at 350 Ah (or two parallel 175 Ah strings)!

Exam Problem 2: Sizing Total Building Power & Transformer

A complete walkthrough of calculating PUE, cooling power, and substation transformers

Problem Statement & Clean Solution Walkthrough

EASY 10 MARKS

- **The Question:** A data centre has 200 server racks drawing 5 kW each. Target PUE is 1.35. Cooling takes 70% of non-IT power. Overall power factor is 0.90. Incorporate a 25% future expansion buffer. Calculate: (a) IT Load kW, (b) Total Facility kW, (c) Cooling kW, (d) Transformer kVA rating, (e) DCiE.
- **Step 1:** IT Load = 200 racks $\times$ 5 kW = 1,000 kW (1.0 Megawatt).
- **Step 2:** Total Building Power = IT Load $\times$ PUE = 1,000 kW $\times$ 1.35 = 1,350 kW.
- **Step 3:** Cooling Power Breakdown:
Non-IT Overhead = 1,350 kW - 1,000 kW = 350 kW.
Cooling Power (70%) = 0.70 $\times$ 350 kW = 245 kW. (Remaining 105 kW goes to lighting and transformer heat losses).
- **Step 4:** Transformer Sizing in kVA:
Building kVA = 1,350 kW / 0.90 Power Factor = 1,500 kVA.
Add 25% future growth margin (multiply by 1.25): 1,500 kVA $\times$ 1.25 = 1,875 kVA.
Standard Engineering Choice: Select standard commercial 2,000 kVA (2 MVA) Dry-Type Transformer! (In Tier IV 2N: install two 2,000 kVA transformers).
- **Step 5:** DCiE Percentage = $(1 / \text{PUE}) \times 100\% = (1 / 1.35) \times 100\% = 74.07\%$!

How to Answer the 3 Most Common Exam Questions

Guaranteed high-scoring answering templates for 10-mark and 15-mark essay questions

Q1: Explain the SLD Diagram (15 Marks)

15-MARK ESSAY

- **How to score full 15 marks:**

1. Draw the Diagram First: Utility Substation -> Transformer -> ATS -> UPS -> Floor PDU -> Overhead Busway -> Rack ePDU -> Server Dual PSUs.
2. Label Path A and Path B: Show two independent paths running side-by-side to prove 2N redundancy.
3. Write 2 lines per component: Briefly explain what RMU does, why transformer has K-factor, why UPS is online double-conversion, and why server has 2 cords.
4. Mention Fault Tolerance: Write: 'If Path A suffers a short circuit, Path B automatically runs 100% of servers with zero downtime.'

Q2: Online Double Conversion UPS (10 Marks)

10-MARK ESSAY

- **How to score full 10 marks:**

1. Draw the 3 Blocks: Rectifier -> DC Bus with Battery -> Inverter. Show the Static Bypass on top and Maintenance Bypass around it.
2. Explain the 3 Modes:
 - Normal Mode: AC converted to DC, then back to clean AC.
 - Outage Mode: Battery instantly feeds inverter with 0 ms delay.
 - Bypass Mode: Fault trips static switch to raw mains in 2 ms.
3. State the Big Benefit: Total electrical isolation, perfect sine wave, 0 ms transfer time (IEC 62040-3 VFI).

Quick-Fire Comparison Tables for 5-Mark Short Notes

Memorize these exact bullet points for rapid scoring in short-note questions

ATS vs. STS Cheat Sheet (5 Marks)

ATS VS STS

- • **Technology:** ATS is mechanical (motor/contactors); STS is silicon electronic (SCR chips).
- • **Speed:** ATS takes 50 to 100 ms; STS takes 2 to 4 ms (sub-cycle).
- • **Server Reboot?:** ATS causes servers to shut down; STS is completely seamless with zero server reboots.
- • **Moving Parts:** ATS has moving parts that wear; STS has zero moving parts.
- • **Location:** ATS goes upstream (Grid vs DG); STS goes downstream (near racks between UPS A & B).

VRLA vs. Lithium-Ion Cheat Sheet (5 Marks)

VRLA VS LI-ION

- • **Lifespan:** VRLA lasts 3-5 years; Lithium lasts 12-15 years.
- • **Weight:** VRLA is extremely heavy (cracks floors); Lithium is 60% lighter.
- • **Space:** VRLA needs huge battery rooms; Lithium takes 70% less footprint.
- • **Heat Tolerance:** VRLA degrades above 25°C; Lithium works fine up to 30°C.
- • **Brain / BMS:** VRLA has dumb cells; Lithium has smart digital BMS monitoring every single cell 24/7.

UNIT 2: FINAL EXAM DAY GOLDEN RULES

Remember these 4 golden habits to secure an A+ in your examination

1. Always Draw

RULE 1

- Even if the question doesn't say 'Draw', always sketch a block diagram.
- Examiners give 50% of the marks just for neat labeled schematics with arrows!

2. Name Drawings

RULE 2

- **Mentioning standards shows real engineering knowledge:**
 - IEEE 1100 (Earthing)
 - ISO 8528-1 (DCC Generators)
 - IEC 62305 (Lightning)
 - IEEE 519 (Harmonics)

3. Write Bold

RULE 3

- Whenever talking about high availability, mention Path A and Path B.
- True Tier IV means 2N dual paths from substation to dual-corded server PSUs.

4. Box Your Answer

RULE 4

- **In numericals:**
 - Write formula clearly.
 - Substitute numbers with units (kW, kVA, Ah).
 - Draw a box around your final answer!